

Kowndinya Boyalakuntla

Third-year Ph.D. student in Computer Science | Rutgers University

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Research interests

I work on learning and planning for robot manipulation and whole-body control. My current research explores action abstractions from human motion capture for planning with learned world models, and Q-guided recovery for frozen imitation policies, toward reliable humanoid loco-manipulation.

Education

Rutgers University Sep 2024 – present

Ph.D. in Computer Science; GPA: 3.85/4.0 Advisor: Prof. Abdeslam Boularias

Rutgers University Sep 2022 – Oct 2024

M.S. in Computer Science; GPA: 3.8/4.0 Advisor: Prof. Abdeslam Boularias

Thesis: Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs.

Indian Institute of Technology Tirupati Jul 2017 – Jul 2021

B.Tech. in Computer Science and Engineering; GPA: 8.42/10.0 Advisors: Prof. Sridhar Chimalakonda, Prof. Kalidas Yeturu

Manuscripts under review

1. PL-MPC: Beyond Policy Alignment: Closing the Planning–Learning Loop for Robot Control with Learned World Models

Kowndinya Boyalakuntla, Yuhan Liu, Abdeslam Boularias

Manuscript, 2026. Under review. [Paper] [Project] [Code]

2. Plan-Conditioned Imitation for Robust Object Retrieval under Self-Occlusion in Dense Clutter

Kowndinya Boyalakuntla, Ajinkya Pawar, Abdeslam Boularias, Jingjin Yu

Manuscript, 2026. Under review. [Paper] [Project] [Code] [Data]

Research experience

Rutgers University Jan 2023 – present

Graduate Student Researcher · New Brunswick, NJ

Conducted research in robot learning for 3D scene understanding, object rearrangement, and placement.

- OVSG (CoRL 2023): Built the evaluation pipeline and DOVE-G benchmark (8 scenarios, 4,000 queries) for context-aware entity grounding; reconstructed 3D scenes and integrated the ConceptFusion baseline.
- LGMCTS (IROS 2024): Co-developed the language-guided rearrangement benchmark with multi-pattern goals and initially infeasible scenes; developed the object-selection network, language parser, and evaluation pipeline.
- DAP (IROS 2024): Collected data and evaluated a diffusion-based affordance prediction framework for multi-modal object storage that combines placeable-region sampling with point-cloud correspondences to estimate 6D placement poses.

RISHA Lab, IIT Tirupati Dec 2019 – May 2022

Undergraduate Student Researcher · Tirupati, India

Developed tools for GitHub repository evaluation and energy-aware mobile application generation.

University of Waterloo Jul 2020 – Sep 2020

Visiting Scholar · Waterloo, Canada

Improved RepoReaper’s recall and reduced false positives in GitHub repository analysis without GHTorrent.

Teaching

Instructor, Rutgers University Fall 2026

CS440: Introduction to Artificial Intelligence Undergraduate course

Selected peer-reviewed publications

1. KARL: Kalman-Filter Assisted Reinforcement Learner for Dynamic Object Tracking and Grasping

Kowndinya Boyalakuntla, Abdeslam Boularias, Jingjin Yu

IROS, 2025 [Paper] [Code]

2. **DAP: Diffusion-based Affordance Prediction for Multi-modality Storage**

Haonan Chang, **Kowndinya Boyalakuntla**, Yuhua Liu, Xinyu Zhang, Liam Schramm, Abdeslam Boularias

IROS, 2024 [\[Paper\]](#) [\[Code\]](#)

3. **LGMCTS: Language-Guided Monte-Carlo Tree Search for Executable Semantic Object Rearrangement**

Haonan Chang, Kai Gao, **Kowndinya Boyalakuntla**, Alex Lee, Baichuan Huang, Harish Udhaya Kumar, Jingjin Yu, Abdeslam Boularias

IROS, 2024 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

4. **Context-Aware Entity Grounding with Open-Vocabulary 3D Scene Graphs**

Haonan Chang, **Kowndinya Boyalakuntla**, Shiyang Lu, Siwei Cai, Eric Pu Jing, Shreesh Keskar, Shijie Geng, Adeeb Abbas, Lifeng Zhou, Kostas Bekris, Abdeslam Boularias

CoRL, 2023 [\[Paper\]](#) [\[Project\]](#) [\[Code\]](#)

Honors & awards

2022 Project showcase, InvenTiv All IITs R&D Fair. Research project advised by Prof. Kalidas Yeturu, showcased at IIT Delhi.

2019 Gold Medal, Inter IIT Technical Meet 8.0. Ashoka's Tech for Change Challenge at IIT Roorkee; high-preparation event among 20 IITs.

Leadership & mentoring

Google Developer Student Clubs, IIT Tirupati

Feb 2019 – Jul 2020

Community Lead

Organized 12 workshops and hackathons for more than 70 students; led 15 mentoring sessions on Flutter and Android.

Professional service & activities

Peer review

- IEEE International Conference on Robotics and Automation (ICRA) (2026)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2025, 2026)
- Conference on Robot Learning (CoRL) (2026)
- The International Journal of Robotics Research (IJRR) (2026)
- International Conference on Automated Planning and Scheduling (ICAPS) (2026)

2021 Developed an Android application for IIT Tirupati's convocation.

2020 Completed 80 hours of National Service Scheme service; developed the IIT Tirupati NSS website and led its maintenance team.